

LTSpice (Lab 03)

ENGR 2405

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Description

This will familiarize the student with the capabilities and basic use of LTSpice which is a freely available circuit simulation and analysis program, similar to many other Simulation Program with Integrated Circuit Emphasis (SPICE) programs (e.g. pSpice, ngSpice, etc).

Calculations

The circuit is solved by the matrix form of Ohm's law, $[I] = [R]^{-1}[V]$, which yields the five mesh currents i_1 through i_5 . The branch currents measured by LTSpice are then recovered from the mesh currents through the following relationships:

$$I(50\ \Omega) = i_1 \quad (1)$$

$$I(20\ \Omega) = i_2 - i_1 \quad (2)$$

$$I(12\ \text{V}) = i_3 \quad (3)$$

$$I(60\ \Omega) = i_5 \quad (4)$$

$$I(80\ \Omega) = i_4 - i_1 \quad (5)$$

Branches on the outer perimeter of the circuit ($50\ \Omega$, $12\ \text{V}$, $60\ \Omega$) belong to a single mesh, so each branch current equals that mesh current directly. The $20\ \Omega$ and $80\ \Omega$ resistors are shared between two adjacent meshes, so the current through each is the difference of the two mesh currents flowing through it. The SciLab program below implements these equations; its runtime printout follows.

```
// Calculations for computing the mesh and branch currents
// ENGR 2405 (ECE 402): Electrical Circuits I
// Abdon Morales

R = [150 -20 0 -80 0;
     -20 65 -30 -15 0;
       0 -30 50 0 -20;
     -80 -15 0 95 0;
       0 0 -20 0 80];

disp("R=");
disp(R);

V = [30;
     0;
    -12;
     20;
    -20];

disp("V=");
disp(V);
```

```

invR = inv(R);
disp("Inverse of R=");
disp(invR);

disp("This is the vector of mesh currents:");
I = invR * V;
disp(I);

disp("This is the vector of branch currents:");
branchCurrents = [I(1);
                  I(2)-I(1);
                  I(3);
                  I(5);
                  I(4)-I(1)];
disp(branchCurrents);

```

Listing 1: SciLab program

```

--> exec('/Users/abdonm/Library/CloudStorage/OneDrive -
TheUniversityofTexasatAustin/Summer 2026/ECE 402/Lab Notes/Lab 03/lab03
.sce', -1)
"R="
150.   -20.    0.   -80.    0.
-20.    65.   -30.   -15.    0.
0.    -30.    50.    0.   -20.
-80.   -15.    0.    95.    0.
0.     0.   -20.    0.    80.
"V="
30.
0.
-12.
20.
-20.
"Inverse of R="
0.0173448   0.0132762   0.0088508   0.0167024   0.0022127
0.0132762   0.0336188   0.0224126   0.0164882   0.0056031
0.0088508   0.0224126   0.0371639   0.0109921   0.009291
0.0167024   0.0164882   0.0109921   0.0271949   0.002748
0.0022127   0.0056031   0.009291    0.002748   0.0148227
"This is the vector of mesh currents:"
0.7039258
0.3470378
-0.1464192
0.8581014
-0.2866048
"This is the vector of branch currents:"
0.7039258
-0.3568879
-0.1464192
-0.2866048
0.1541756

```

Listing 2: SciLab runtime output

Simulations

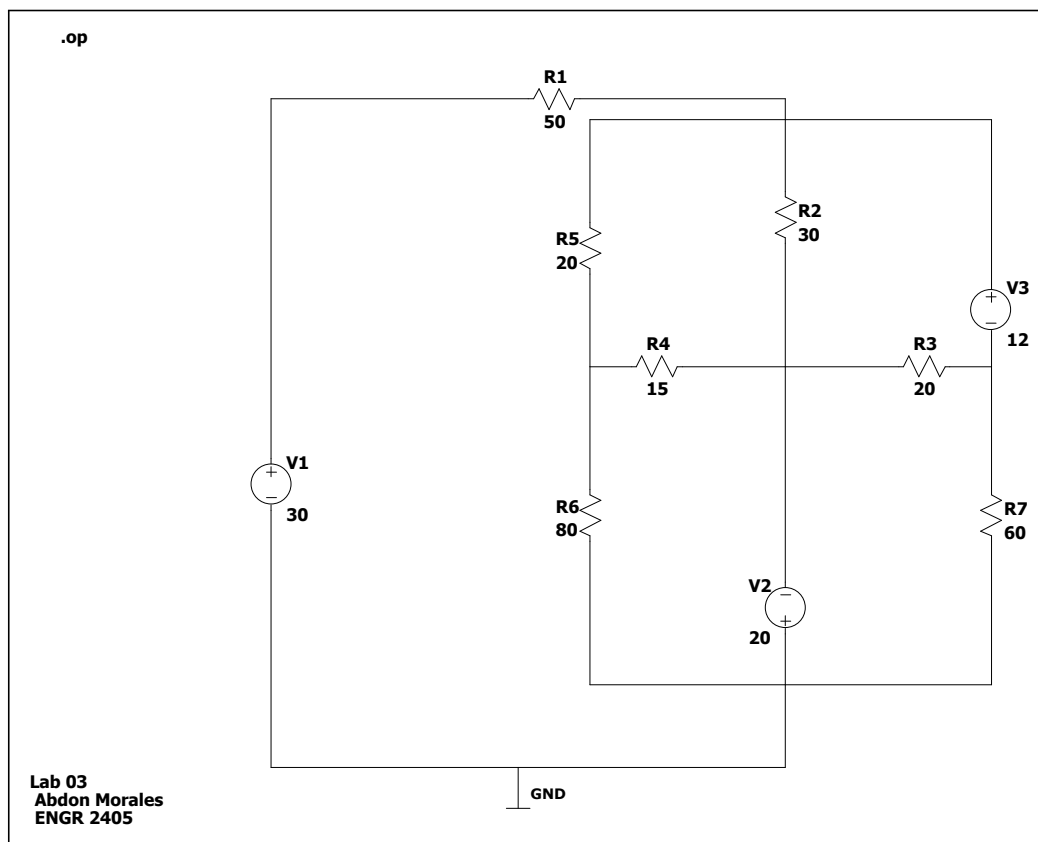


Figure 1: LTSpice schematic

```
* /Users/abdonm/Library/CloudStorage/OneDrive-TheUniversityofTexasatAustin
  /Summer 2026/ECE 402/Lab Notes/Lab 03/Lab03-Morales-Abdon.asc
* Generated by LTspice 26.0.2 for MacOS.
V1 N001 0 30
V2 0 N004 20
V3 N002 N005 12
R1 N002 N001 50
R2 N002 N004 30
R3 N005 N004 20
R4 N004 N003 15
R5 N003 N002 20
R6 0 N003 80
R7 N005 0 60
.op
* Lab 03\nAbdon Morales\nENGR 2405
.backanno
.end
```

Listing 3: LTSpice netlist

Results

The DC operating point simulation produced the following node voltages and device currents:

--- Operating Point ---

```
V(n001):    30          voltage
V(n002):   -5.19629     voltage
V(n003):   -12.334      voltage
V(n004):   -20          voltage
V(n005):  -17.1963      voltage
I(R1):     -0.703926     device_current
I(R2):      0.493457     device_current
I(R3):      0.140186     device_current
I(R4):     -0.511064     device_current
I(R5):     -0.356888     device_current
I(R6):      0.154176     device_current
I(R7):     -0.286605     device_current
I(V1):     -0.703926     device_current
I(V2):     -1.14471      device_current
I(V3):     -0.146419     device_current
```

Observations & Discussion

The branch currents computed in SciLab were compared against the LTSpice DC operating point simulation results, using the mesh-to-branch reconciliation equations given in the Calculations section. Evaluating those equations from the SciLab mesh currents and comparing to the LTSpice device currents gives:

Branch	SciLab (A)	LTSpice (A)	Agreement
$I(50\ \Omega)$	+0.7039	-0.7039	magnitude only (sign flipped)
$I(20\ \Omega)$	-0.3569	-0.3569	exact
$I(12\ \text{V})$	-0.1464	-0.1464	exact
$I(60\ \Omega)$	-0.2866	-0.2866	exact
$I(80\ \Omega)$	+0.1542	+0.1542	exact

Four of the five branch currents agree exactly, including sign. The only difference appears in the $50\ \Omega$ resistor (R1), where the magnitude is identical (0.7039 A) but the sign is reversed.

This discrepancy is a sign-convention difference, not a numerical error. LTSpice reports the current through a two-terminal device as flowing *into the first node listed for that device in the netlist*. The netlist entry R1 N002 N001 therefore defines positive current as flowing from N002 to N001 inside R1. The matrix computation instead defines $I(50\ \Omega) = i_1$ using the mesh-current reference direction, which is opposite to the netlist node ordering for this branch. The two results describe the same physical current with opposite reference directions, so they differ only in sign. The remaining four branches happen to have netlist node orderings that agree with their mesh reference directions, which is why their signs match directly.

The node voltages are also internally consistent: $V(\text{n001}) = 30\ \text{V}$ corresponds to the positive terminal of the 30 V source on the far-left node, with the bottom node taken as ground (0 V). The simulation therefore confirms both the topology and the element values of the circuit.